

Consumers' Valuation of Rice-Grade Labeling

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The current Korean rice-grading system has a “no test” option that allows rice to not be graded in the market. This study examines Korean consumers’ valuation of a rice-grading system without the “no test” option. We apply a nonhypothetical experimental auction to elicit consumers’ willingness to pay for each rice grade and identify the impact of the provision of additional grading information on product valuation. We then use contingent and inferred valuations to obtain consumers’ valuation of a mandatory rice-grading system without the “no test” option. We find that Korean consumers are willing to pay an additional premium for each rice grade and that rice-grading information is the most important factor that differentiates the rice products. Rice consumers in Korea also strongly prefer a mandatory rice-grading system without the “no test” option.

L'actuel système coréen de classement du riz comporte une option « exempt de test » qui permet au riz de ne pas être classé pour le marché. Cet article examine l'évaluation par le consommateur d'un système de classement du riz sans option d'exemption de test. Nous appliquons une enchère expérimentale réelle afin d'obtenir la volonté de payer du consommateur pour chaque catégorie de riz et d'identifier l'impact d'informations supplémentaires au sujet de la classification sur l'évaluation du produit. Nous utilisons ensuite les évaluations contingente et inférée pour obtenir l'évaluation par le consommateur d'un système de classification du riz sans l'option « exempt de test ». Nous constatons que les consommateurs coréens sont disposés à payer plus pour chaque catégorie de riz et que les informations sur la classification du riz représentent le plus important facteur de différenciation des produits de riz. Les consommateurs coréens de riz préfèrent aussi fortement le système de classification du riz ne comportant pas l'option « exempt de test ».

INTRODUCTION

Rice is the main staple food in Korea and is a major source of income for Korea's farmers. Approximately 47% of Korean's caloric intake and 70% of farm income come from rice consumption and production, respectively (Agriculture, Food and Rural Affairs Statistics Yearbook 2016; National Food and Nutrition Statistics 2016). Despite the prominence of rice in Korea, there was an absence of a standard rice-grading system until 2011. So,

for a long time, consumers had no information about the quality grade of the rice that they purchased. Consumers also had difficulty distinguishing the quality of domestic rice from that of the imported rice in the market. Recently, Lee et al (2014) found that Korean consumers prefer the quality of imported rice over that of domestic rice when they are not given information about the country of origin and are only allowed to taste the rice. However, they value the domestic rice more than the imported rice when given information about the country of origin. The amount of imported rice has also increased and has triggered more market competition since the opening up of the Korean rice market through a minimum market access (MMA) scheme during the 1995–2014 period and the reform of the rice import framework that transitioned it into a tariffication system in 2015 (Lee et al 2014; Kim et al 2015). Thus, it is imperative that the Korean rice industry distinguishes the quality of its domestic rice from that of imported rice in the market.

The Korean government introduced a new domestic rice-grading system in 2011. The purpose of the system was to satisfy consumers' right to information and improve the quality of domestic rice. Based on this new system, Korean domestic rice was classified into five grades or ranks (with the first rank representing the best quality rice). The government required that rice sold in retail markets be labeled with grading information so that consumers could make informed purchasing decisions at the time of purchase. However, this mandatory rice-grading system caused an additional marketing cost for rice manufacturers because they had to purchase extra equipment to conduct the rice-grading test. The complex five-rank system also confused rice manufacturers and caused many grading violations, which resulted in the government imposing penalties on rice manufacturers. The cost burden and strict regulations of the new, mandatory grading system led to most rice manufacturers resisting the policy. As a result, the Korean government revised the rice-grading system in 2013 to make it less restrictive to rice manufacturers. Specifically, the revised system classified domestic rice into three grades instead of five: "super," "good," and "normal."¹ Importantly, the new grading system now allowed a "no test" option (which means that the rice has not been tested for grading) in order to make the policy more flexible for domestic rice manufacturers.

The revised rice-grading system, however, has not proven to be effective because allowing the "no test" option does not provide a strong incentive for rice manufacturers to follow the policy. According to a recent investigation of the National Agricultural Products Quality Management Service (NAQS), approximately 75% of domestic rice for sale in retail markets was labeled "no test" (The Agriculture Policy Focus 2016). With such a high percentage of rice labeled "no test," consumers cannot always make informed purchasing decisions in the market, a situation that has caused low consumer trust of domestic rice products (Lee et al 2015b). There are a number of possible reasons for the high percentage of rice bearing a "no test" label. First, the price of rice in Korea depends heavily on the brand and the region of production. Thus, rice manufacturers may have less incentive to consider the quality of the rice to obtain an additional premium. Another possible reason for the high percentage of "no test" rice is the cost burden of conducting

¹ The classification of the quality grades in the current grading system is based on the maximum content criteria of moisture, broken rice, fracture particles, damaged grain, heat-loss grain, and foreign materials.

a rice-grading test, which involves the purchase of equipment to perform testing and the hiring of more employees.

To make the grading system effective, policymakers need to consider disposing of the “no test” option in the current rice-grading system. To determine the market feasibility of implementing a new mandatory rice-grading system without the “no test” option, however, policymakers first need to identify the premium that consumers are willing to pay for each rice quality grade (i.e., “super,” “good,” and “normal” grades) compared to rice that has a “no test” label, and these quality grade premiums need to be explained to rice manufacturers to incentivize them to comply with the system. It is also necessary to determine whether consumers’ willingness to pay (WTP) for a mandatory grading system without the “no test” option can partially support the cost of implementing such a system. Hence, the objectives of this study are to (i) identify rice consumers’ WTP for each rice quality grade (i.e., “super,” “good,” and “normal” grades) compared to rice that has a “no test” label, and (ii) investigate consumers’ valuation of a mandatory rice-grading system without the “no test” option. Since it would be beneficial as well for policymakers to know consumers’ responses to rice-grading information under the mandatory rice-grading system without the “no test” option, we also analyze the effects of the provision of additional grading information on consumers’ WTP for each rice quality grade. Despite the important role of the quality grading system in the Korean rice industry, no other study has examined this topic in Korea. Furthermore, to our knowledge, this study is the first to focus on the importance of a standard rice-grading system in the domestic market of a major rice importing country.

We utilized three valuation methods in this study. First, we used a nonhypothetical experimental auction (i.e., the random n th price auction) to elicit consumers’ WTP for each rice quality grade. We then applied the contingent valuation (CV) and inferred valuation (IV) methods (i.e., a double-bounded dichotomous choice [DBDC] question) to obtain consumers’ valuation of a mandatory rice-grading system without the “no test” option. Finally, in order to investigate the potential effects of rice-grading information, we also tested three types of information about each rice quality grade: no information, grade-only information, and detailed grade information.

EXPERIMENTAL DESIGN AND METHOD

The main subjects of our experiment were housewives in their twenties to sixties; this demographic represents the primary rice shoppers in Korea (Lee et al 2014). A total of 212 participants living in Seoul, the capital city of Korea, participated in our auction experiment. Of the participants, 103 were recruited through a research company; the others (109) were directly recruited by the researchers. We considered the participants’ ages and regions when recruiting them. The age and regional distribution of the sample is representative of Seoul’s population, as shown in Table 1. According to the local census in 2015, approximately 5.11 million women live in Seoul, and approximately 3.77 million women are aged between 20 and 60. The age distribution in our sample is similar to the local census. The one-sample proportion tests also indicate that we fail to reject the hypothesis that the proportions of age distributions in our sample are similar to the known proportions of age distributions in the local census. Although the regional proportions in our sample slightly differ from that of the local census, the proportion tests indicate

Table 1. Age and regional distribution of participants ($N = 212$)

Variable	Categories	Proportion of population (%)	Proportion of sample (%)	Equality test of proportion
Age	20s	18.9	13.3	$z = -0.8$ (p -value = 0.4)
	30s	22.2	22.3	$z = 0.1$ (p -value = 0.9)
	40s	22.8	21.3	$z = -0.3$ (p -value = 0.8)
	50s	22.1	24.1	$z = 0.4$ (p -value = 0.7)
	60s	14.1	18.9	$z = 0.9$ (p -value = 0.4)
Region	East (North)	31.5	27.8	$z = -0.7$ (p -value = 0.5)
	West (North)	17.1	21.7	$z = 0.7$ (p -value = 0.5)
	East (South)	30.4	24.1	$z = -0.9$ (p -value = 0.4)
	West (South)	21.0	26.4	$z = 0.9$ (p -value = 0.4)

that we cannot reject the hypothesis that the proportions of regional distribution in our sample are similar to the proportions in the local census.

Valuation of Rice Grades

We first investigated the participants' valuation of each rice grade using a nonhypothetical experimental auction. We utilized the random n th price auction mechanism since it is incentive compatible and widely used by many researchers (List 2003; Parkhurst et al 2004; Lee et al 2011, 2014). In the random n th price auction, the number of winners is determined after the bidding process through a random drawing of a number between 2 and n (total number of participants in the session). The number of winners is then the top $(n-1)$ bidders in the session. Each winner gets the product being auctioned and pays the n th highest bid in the session. In our experiment, we also applied the full bidding approach, whereby participants were asked to bid simultaneously on four different rice products: "super," "good," "normal," and "no test." A binding rice was then randomly chosen at the end of the bidding process. More details of the auction procedures are discussed below.

The experiment was conducted from May 27 to July 8, 2015. A total of 16 sessions were conducted, with each group consisting of 10–15 subjects. Each participant received 15,000 KRW (approximately 17 CAD)² as a participation fee in the experiment, which lasted approximately 50 minutes on average. The auctioned products were 1 kg packs of the same brand of rice produced in Hwaseong, in the Gyeonggi province of Korea. In the experiment, four different grades of rice were used: "super," "good," "normal," and "no test." The rice products were uniformly packed using clear plastic bags and were placed on the subjects' desks.

The experiment was composed of three bidding rounds. Each round differed in the amount of information given to subjects about the rice grades. No information was provided in the first round. In the second round, participants were provided with simple rice-grading information (from the labels of each rice grade). They were then provided

² KRW and CAD represent Korean Won and Canadian Dollar, respectively. 1 CAD is about 900 KRW.

Table 2. Information rounds in auction experiment

Round	Round 1	Round 2	Round 3
Information supplied	No grade information; only visual, touch, and smell test	Simple grade information; visual, touch, and smell test	Full detailed information on the meaning of each rice grade; visual, touch, and smell test

with additional detailed information about rice grades in the third round (the information came from the labels and from a supplied, detailed explanation of the meaning of each grade). We also allowed participants to see, touch, and smell each rice sample in all rounds. Table 2 shows the three bidding rounds of the experiment.

Upon arrival, participants received an ID number and were assigned seats. They were also required to sign a consent form. After completing the consent form, participants were given detailed instructions by the moderator about how the experimental auction would work. Notably, the participants were advised that the best strategy is to submit a bid that is equal to their true valuation of each rice product. Before the actual rice auction, we conducted a practice auction with the participants using cereal bars. This practice session helped them understand the nature of the auction and become more familiar with the auction procedure. Then, prior to the rice auctions, participants were asked a simple question to determine if they fully understood the auction procedures.

Following the practice auction, participants were informed that there would be three rounds of auction. They were also told that a binding round and binding rice would be randomly selected at the end of the three rounds. In round 1, participants first examined the four different blinded rice products placed on their desks (i.e., visually, by touch, and by smell). After this examination, participants submitted sealed bids representing their WTP to buy each of the four different types of rice. In round 2, participants were given the quality grade labels of each rice. After receiving the labels, participants then submitted their bids for each rice. In round 3, participants were given additional detailed information about the meaning of each rice grade and then submitted bids for each rice.

Valuation of a Mandatory Rice-Grading System

As previously discussed, we also conducted a CV and IV study with the same participants after they had completed the auction experiment, in order to elicit their valuation of a mandatory rice-grading system without the “no test” option. We used two stated preference valuation questions to compare results and test the robustness of results from two valuation methods. For the CV method, participants were directly asked what they were willing to pay, while for the IV method, they were asked what they thought an average person would be willing to pay. CV is one of the most popular methods used for valuing nonmarket goods; however, individuals' valuations may suffer from hypothetical bias because of the hypothetical nature of the question.

One of the potential reasons for hypothetical bias is social desirability bias. Thus, participants who answer a direct question in the presence of an interviewer may present socially acceptable preferences instead of showing their true preferences (Leggett et al

A1. The average retail price of 1 kg of rice in Korea is 4,000 KRW. If the Korean government decides to dispose of the “no test” option in the current rice-grading system, would you be willing to pay an additional () KRW above the average price of 4,000 KRW/kg?

- ① Yes (Go to Question 1-1) ② No (Go to Question 1-2)

1-1. If so, would you be willing to pay an additional _____ KRW?

- ① Yes ② No Upper ($\times 2$) price

1-2. If so, would you be willing to pay an additional _____ KRW?

- ① Yes ② No Lower ($\times 0.5$) price

Figure 1. Example of double-bounded dichotomous choice question

2003; List et al 2004). IV was originally developed to alleviate potential social desirability bias by asking indirect questions of participants (Lusk and Norwood 2009; Stachtiaris et al 2013; Yadav et al 2013). Participants are asked to predict other people’s potential behaviors, and thus, infer their preferences. This indirect questioning can free participants from moral considerations, and consequently, minimize social desirability bias. Studies have also concluded that IV moderates social desirability and generates lower WTP values than CV (Lusk and Norwood 2009; Yadav et al 2013).

In the CV and IV surveys, we provided all subjects with detailed and clear information about the current rice-grading system and the possibility of removing the “no test” option in the current system. We also provided information about why consumers might care about the issue of the “no test” option in the current system and about a potential increase in the price of domestic rice after removing the “no test” option. Each subject was then asked the CV and IV WTP questions about the mandatory rice-grading system without the “no test” option. In order to take the possible ordering effect into account, half of the subjects (in eight sessions) were randomly asked to answer the CV question first, followed by the IV question; the remaining subjects were asked to answer in reverse order.

The DBDC approach was used as the format of the WTP questions in both the CV and IV methods since this approach is statistically more efficient than a single-bounded approach (Hanemann et al 1991; Mergenthaler et al 2009; Holmquist et al 2012). Participants were asked the following question in the CV survey: “The average retail price of 1 kg of rice in Korea is 4,000 KRW (approximately 4.4 CAD). If the Korean government decides to dispose of the ‘no test’ option in the current rice-grading system, would you be willing to pay an additional (X) KRW above the average price of 4,000 KRW/ kg?” Based on their responses, participants were asked a follow-up question. If a respondent answered “yes” to the first question, she was asked whether she would be willing to pay double the amount proposed in the first question. If a respondent answered “no” to the first question, she was asked whether she would be willing to pay half the amount indicated in the first question (see Figure 1). For example, if a respondent was asked whether she would be willing to pay an additional 200 KRW in the first question and she responded with a “yes” (“no”) to the question, she was then asked whether she would be willing to pay an additional 400 KRW (100 KRW) in the follow-up question.

In the IV survey, participants were asked the following question: “The average retail price of 1 kg of rice in Korea is 4,000 KRW (approximately 4.4 CAD). If the Korean government decides to dispose of the ‘no test’ option in the current rice-grading system, do you think an average consumer would be willing to pay an additional (X) KRW above the average price of 4,000 KRW/kg?” The procedure used for the follow-up question was the same as that of the CV survey. To avoid a starting-point bias, we randomly selected a first bid value from the predetermined price distribution (10 levels of prices from 200 to 2,000 KRW in increments of 200 KRW) for each respondent. It should be noted that we had conducted a pilot survey to construct the price distribution method used in this study. In the pilot survey, we used the same CV and IV questions with another price distribution that had a larger price range (12 price levels). Based on answers from respondents in the pilot survey, we then revised the upper and lower values from the original price distribution. For the actual survey, a cheap talk script was read and shown to participants (see the Appendix) prior to the CV and IV WTP questions to potentially reduce hypothetical bias (see the discussion about the cheap talk in Cummings and Taylor 1999; Lusk 2003; Silva et al 2011).

To analyze the DBDC data from the CV and IV, we assumed that consumer's WTP is influenced by participants' observable characteristics (i.e., age, education level, income, and household size), rice consumption patterns (i.e., frequency at which they check rice grades, frequency at which they purchase rice, and purchasing price), their desire for rice grading, and their prior awareness about rice grading. Given this assumption, the WTP for the mandatory rice-grading system can be specified as

$$WTP_i = x_i \beta + \varepsilon_i \quad (1)$$

where WTP_i is individual i 's unobservable true WTP; x_i is a vector of socioeconomic factors, rice consumption patterns, the desire for rice grading, and prior awareness about rice grading; β is the vector of parameter estimates; and ε_i is an i.i.d. component.

Given our use of the DBDC format, we have four possible response cases in both CV and IV: (yes, no), (yes, yes), (no, yes), and (no, no) (Hanemann et al 1991). The probability of each of the four cases can be provided by

$$Bid_i^1 = 1 \text{ (yes)} \quad \text{and} \quad Bid_i^2 = 0 \text{ (no)} \quad (2)$$

$$\begin{aligned} \Pr(Bid_i^1 = 1, Bid_i^2 = 0 | x_i) &= \Pr(t^1 \leq x_i' \beta + \varepsilon_i < t^2) \\ &= \Phi\left(x_i' \frac{\beta}{\sigma} - \frac{t^1}{\sigma}\right) - \Phi\left(x_i' \frac{\beta}{\sigma} - \frac{t^2}{\sigma}\right) \end{aligned}$$

where Bid_i^1 and Bid_i^2 are the responses to the first and second questions in the DBDC, respectively; $\Pr(Bid_i^1 = 1, Bid_i^2 = 0 | x_i)$ is the conditional probability that an individual answers “yes” to the first question and “no” to the second question; t^1 and t^2 are the first and second bid amounts, respectively.

$$Bid_i^1 = 1 \text{ (yes)} \quad \text{and} \quad Bid_i^2 = 1 \text{ (yes)} \quad (3)$$

$$\begin{aligned}\Pr(Bid_i^1 = 1, Bid_i^2 = 1 | x_i) &= \Pr(x_i' \beta + \varepsilon_i > t^1, x_i' \beta + \varepsilon_i \geq t^2) \\ &= \phi\left(x_i' \frac{\beta}{\sigma} - \frac{t^2}{\sigma}\right)\end{aligned}$$

where $\Pr(Bid_i^1 = 1, Bid_i^2 = 1 | x_i)$ is the conditional probability that an individual answers “yes” to both the first and second questions.

$$Bid_i^1 = 0 \text{ (no)} \quad \text{and} \quad Bid_i^2 = 1 \text{ (yes)} \quad (4)$$

$$\begin{aligned}\Pr(Bid_i^1 = 0, Bid_i^2 = 0 | x_i) &= \Pr(t^2 \leq x_i' \beta + \varepsilon_i < t^1) \\ &= \phi\left(x_i' \frac{\beta}{\sigma} - \frac{t^2}{\sigma}\right) - \phi\left(x_i' \frac{\beta}{\sigma} - \frac{t^1}{\sigma}\right)\end{aligned}$$

where $\Pr(Bid_i^1 = 0, Bid_i^2 = 1 | x_i)$ is the conditional probability that an individual answers “no” to the first question and “yes” to the second question.

$$Bid_i^1 = 0 \text{ (no)} \quad \text{and} \quad Bid_i^2 = 0 \text{ (no)} \quad (5)$$

$$\begin{aligned}\Pr(Bid_i^1 = 0, Bid_i^2 = 1 | x_i) &= \Pr(x_i' \beta + \varepsilon_i < t^1, x_i' \beta + \varepsilon_i < t^2) \\ &= 1 - \phi\left(x_i' \frac{\beta}{\sigma} - \frac{t^2}{\sigma}\right)\end{aligned}$$

where $\Pr(Bid_i^1 = 0, Bid_i^2 = 0 | x_i)$ is the conditional probability that an individual answers “no” to both the first and second questions. Based on the probability of each of the four response cases, we can construct a likelihood function to obtain parameter estimates using maximum likelihood estimation. The likelihood function that needs to be maximized is then

$$\begin{aligned}\sum_{i=1}^N \pi_i^{yn} \ln\left(\phi\left(x_i' \frac{\beta}{\sigma} - \frac{t^1}{\sigma}\right) - \phi\left(x_i' \frac{\beta}{\sigma} - \frac{t^2}{\sigma}\right)\right) &+ \pi_i^{yy} \ln\left(\phi\left(x_i' \frac{\beta}{\sigma} - \frac{t^2}{\sigma}\right)\right) \\ + \pi_i^{ny} \ln\left(\phi\left(x_i' \frac{\beta}{\sigma} - \frac{t^2}{\sigma}\right) - \phi\left(x_i' \frac{\beta}{\sigma} - \frac{t^1}{\sigma}\right)\right) &+ \pi_i^{nn} \ln\left(1 - \phi\left(x_i' \frac{\beta}{\sigma} - \frac{t^2}{\sigma}\right)\right)\end{aligned} \quad (6)$$

where $\pi_i^{yes,no}$, $\pi_i^{yes,yes}$, $\pi_i^{no,yes}$, and $\pi_i^{no,no}$ are indicator variables that take the value of one or zero depending on individuals' responses.

EXPERIMENTAL RESULTS

Summary statistics for the socioeconomic characteristics of our samples are presented in Table 3. The average age of participants is approximately 46 years, and the average number

Table 3. Participants' socioeconomic characteristics ($N = 212$)

Variable	Categories	Mean	SD
Age	Years	45.8	13.4
Household size	Persons	3.3	1.2
PFrequency ^a	Times/year	8.2	5.9
Need ^b	1: Strongly disagree—5: Strongly agree	4.1	0.8
Awareness ^c	1: Not at all—4: A great deal	1.8	0.7
Monthly income (unit: 1,000 KRW)	Less than KRW 2,990	17.2%	
	KRW 3,000–6,990	63.2%	
	More than KRW 7,000	19.6%	
	Education level		
Education level	High school graduate	35.4%	
	College graduate	59.9%	
	Graduate degree	4.7%	
	GFrequency ^d		
GFrequency ^d	Never	26.9%	
	Less than 20%	13.7%	
	20–40%	18.4%	
	40–60%	9.9%	
	60–80%	14.2%	
	Every purchase	16.9%	
	PPrice ^e (unit: 1,000 KRW)		
	Less than KRW 38	18.4%	
PPrice ^e (unit: 1,000 KRW)	KRW 39–51	50.5%	
	More than KRW 52	31.1%	

^aFrequency of purchasing rice products per year.

^bRice grading is needed for rice consumers' right to information.

^cThe level of prior awareness of rice grading.

^dFrequency of checking rice grades when purchasing rice.

^ePurchasing price level of rice (20 kg).

of family members in each household is 3.3 persons. According to the income categories, 63.2% of the participants have an average monthly household income between 3 million KRW and 6.99 million KRW, and 19.6% of the participants have an income level of more than 7 million KRW. Almost all participants are either high school or college graduates. On average, participants purchase rice approximately eight times a year. Furthermore, more than 50% of participants buy rice with a price level between 39 thousand KRW and 51 thousand KRW per 20 kg. On average, participants decidedly agree that they need a standard rice-grading system to enhance consumers' right to information (4.1 on a 1 to 5 scale, where 1 = "strongly disagree" to 5 = "strongly agree"). Their prior awareness of the rice-grading system is, however, quite low (1.8 on a 1 to 4 scale, where 1 = "not at all" to 4 = "a great deal"). In addition, only about 17% of participants check the grade level every time they purchase rice.

Objective 1: Consumers' Valuation of Each Rice Grade and Effect of Grade Information

Table 4 shows the bid distributions across the information rounds. Focusing initially on the first round (with no information on the quality grade and just evaluating by visual examination, touch, and smell), the mean bid for the "super" grade has the highest value

Table 4. Bid distribution across information rounds ($N = 212$, KRW/kg)

Grade	Description	Information treatment		
		No information (Round 1)	Grade only (Round 2)	Grade in detail (Round 3)
Super	Mean	3,795	4,392	4,338
	Median	3,900	4,300	4,200
	SD	843	940	962
Good	Mean	3,760	3,958	3,919
	Median	3,900	4,000	4,000
	SD	708	741	814
Normal	Mean	3,660	3,548	3,521
	Median	3,800	3,800	3,700
	SD	727	780	822
No test	Mean	3,700	3,209	3,160
	Median	3,800	3,500	3,500
	SD	832	905	927

at 3,795 KRW/kg (approximately 4.2 CAD), followed by the “good,” “no test,” and “normal” grades. However, the analysis of the variance (ANOVA) test shows that we fail to reject the hypothesis of similar values among the rice grades and “no test” (see Table 5). The mean equality tests between each grade and “no test” in the first round also show that bid differences are not statistically significant in all cases,³ indicating that rice consumers cannot easily distinguish the quality of different rice grades without any information about the quality grade.

Participants’ bids for each rice grade dramatically change in the second round (with grade only information). The mean bid for the “super” grade is the highest value at 4,392 KRW/kg (approximately 4.9 CAD), followed by the “good,” “normal,” and “no test” rice. Interestingly, the mean bids for the “super” and “good” grades increase by 15.7 % and 5.3 %, respectively, from the mean bids in the first round. The mean bids for the “normal” grade and “no test” rice decrease by 3.2% and 15.3%, respectively. An ANOVA test also indicates that we reject the hypothesis of similar values among the grades and “no test” in the second round (see Table 5). The mean bid differences between the grades and “no test” are also statistically significant in all cases, suggesting that rice consumers are willing to pay an additional premium for each rice grade compared to the “no test” rice when given rice-grading information (a 36.9% premium for the “super” grade, a 23.3% premium for the “good” grade, and a 10.6% premium for the “normal” grade). In the third round, with participants receiving detailed information about the meaning of the rice grades, the bids for each rice grade slightly decrease compared to the bids in the second round. However, the mean bid differences between rounds 2 and 3 are not statistically significant (see Table 6), indicating that participants still bid higher for each rice grade than the “no test” rice when given detailed information about rice grades.

³ We applied the Bonferroni multiple comparison test to compare mean bids following the suggestion of a reviewer.

Table 5. Bid comparisons across rice grades ($N = 212$, KRW/kg)

Information treatment	Grade comparison	Mean difference
No information (Round 1)	$H_0 : \overline{WTP}_{Super} - \overline{WTP}_{Good} = 0$	34.6
	$H_0 : \overline{WTP}_{Super} - \overline{WTP}_{Normal} = 0$	134.7
	$H_0 : \overline{WTP}_{Super} - \overline{WTP}_{No\ test} = 0$	94.2
	$H_0 : \overline{WTP}_{Good} - \overline{WTP}_{Normal} = 0$	100.0
	$H_0 : \overline{WTP}_{Good} - \overline{WTP}_{No\ test} = 0$	59.6
	$H_0 : \overline{WTP}_{Normal} - \overline{WTP}_{No\ test} = 0$	-40.5
ANOVA	F -statistic	1.3
Grade only (Round 2)	$H_0 : \overline{WTP}_{Super} - \overline{WTP}_{Good} = 0$	433.5***
	$H_0 : \overline{WTP}_{Super} - \overline{WTP}_{Normal} = 0$	843.6***
	$H_0 : \overline{WTP}_{Super} - \overline{WTP}_{No\ test} = 0$	1,182.1***
	$H_0 : \overline{WTP}_{Good} - \overline{WTP}_{Normal} = 0$	410.1***
	$H_0 : \overline{WTP}_{Good} - \overline{WTP}_{No\ test} = 0$	748.6***
	$H_0 : \overline{WTP}_{Normal} - \overline{WTP}_{No\ test} = 0$	338.5***
ANOVA	F -statistic	77.6***
Grade in detail (Round 3)	$H_0 : \overline{WTP}_{Super} - \overline{WTP}_{Good} = 0$	419.5***
	$H_0 : \overline{WTP}_{Super} - \overline{WTP}_{Normal} = 0$	817.3***
	$H_0 : \overline{WTP}_{Super} - \overline{WTP}_{No\ test} = 0$	1,178.4***
	$H_0 : \overline{WTP}_{Good} - \overline{WTP}_{Normal} = 0$	397.8***
	$H_0 : \overline{WTP}_{Good} - \overline{WTP}_{No\ test} = 0$	758.9***
	$H_0 : \overline{WTP}_{Normal} - \overline{WTP}_{No\ test} = 0$	361.1***
ANOVA	F -statistic	70.1***

Note: Values in the column “Mean difference” represent the mean bid differences between the grades in each information treatment, and *** signifies that the mean differences are statistically significant at the 1% significance level.

We further investigate the effect of sequential provision of information about quality grade on participants' valuation of each rice grade. Table 6 shows the test results from both the ANOVA test among the three information rounds and the mean equality tests across the rounds. The tests show that we reject the hypothesis of similar values among the three information rounds for each of “super” grade, “good” grade, and “no test,” but we cannot reject the hypothesis for “normal” grade. According to the mean equality tests between the information rounds, participants bid significantly higher for both the “super” and “good” grades when provided with either grade-only information or detailed grade information in rounds 2 and 3, respectively, than when provided with no grade information. Although participants respond more sensitively to grade-only information compared to detailed grade information, these differences in bids across the rice grades are not statistically significant.

However, we observe that the average bids for both the “normal” grade and “no test” rice decrease after the provision of either grade-only or detailed grade information. These information effects are statistically significant for “no test” rice, but they are not statistically significant for “normal” grade rice.

Since the descriptive statistics and related unconditional tests cannot completely reveal the value differences across the grades and the effects of grading information across

Table 6. Bid comparisons across information rounds ($N = 212$, KRW/kg)

Grade	Comparison of information treatment	Mean difference
Super	$H_0 : \overline{WTP}_{GradeOnly} - \overline{WTP}_{NoInfo} = 0$	596.9***
	$H_0 : \overline{WTP}_{GradeDetail} - \overline{WTP}_{NoInfo} = 0$	543.8***
	$H_0 : \overline{WTP}_{GradeDetail} - \overline{WTP}_{GradeOnly} = 0$	-53.1
ANOVA	F -statistic	27.6***
Good	$H_0 : \overline{WTP}_{GradeOnly} - \overline{WTP}_{NoInfo} = 0$	198.0**
	$H_0 : \overline{WTP}_{GradeDetail} - \overline{WTP}_{NoInfo} = 0$	158.9*
	$H_0 : \overline{WTP}_{GradeDetail} - \overline{WTP}_{GradeOnly} = 0$	-39.2
ANOVA	F -statistic	4.1**
Normal	$H_0 : \overline{WTP}_{GradeOnly} - \overline{WTP}_{NoInfo} = 0$	-112.1
	$H_0 : \overline{WTP}_{GradeDetail} - \overline{WTP}_{NoInfo} = 0$	-138.9
	$H_0 : \overline{WTP}_{GradeDetail} - \overline{WTP}_{GradeOnly} = 0$	-26.8
ANOVA	F -statistic	1.9
No test	$H_0 : \overline{WTP}_{GradeOnly} - \overline{WTP}_{NoInfo} = 0$	-491.0***
	$H_0 : \overline{WTP}_{GradeDetail} - \overline{WTP}_{NoInfo} = 0$	-540.5***
	$H_0 : \overline{WTP}_{GradeDetail} - \overline{WTP}_{GradeOnly} = 0$	-49.4
ANOVA	F -statistic	23.9***

Note: Values in the column “Mean difference” represent the mean bid differences between the information treatments, and *, **, and *** signify that the mean differences are statistically significant at 10%, 5%, and 1% significance levels, respectively.

the rounds, we further examined the premium differences across the grades and the effects of information on the premium for each rice grade (the differences in WTP between each rice grade and the “no test” rice) at the individual level by estimating random effects panel regression models. For the regression analysis, we withdrew 20 individuals because these participants had not completed some important questions that were used in the regression as variables. Thus, 192 respondents were used in the regression analysis. To evaluate whether there is an information effect across the rounds, the models first included dummy variables for round 2 (grade-only information) and round 3 (detailed grade information). Since product valuations can be affected by individual characteristics, the models also included participants’ socioeconomic characteristics (i.e., age, education, income, and household size), the frequency at which individuals check rice grades when purchasing rice (GFrequency), the frequency at which rice is purchased per year (PFrequency), the purchasing price of rice (PPrice), participants’ level of prior awareness about the rice-grading system (Awareness), and participants’ desire for a rice-grading system to enhance consumers’ right to have information (Need).

Columns (1)–(3) of Table 7 show the estimation results from the random effects regression models for the premiums of each rice grade. All intercepts in the three models are positive and statistically significant, indicating that each rice grade commands an additional premium compared to the “no test” rice. The provision of both grade-only and detailed grade information positively affect the premium for each rice grade and have bigger impacts on the “super” grade, followed by the “good” and “normal” grades, a finding that is consistent with prior unconditional test results. We also combined data

Table 7. Random effects regression models of quality grade premium

Variable	Premium for super Coefficient (Standard error)	Premium for good Coefficient (Standard error)	Premium for normal Coefficient (Standard error)	Pooled Coefficient (Standard error)
Round2	1,096.0 (80.3)***	692.6 (64.0)***	392.4 (64.1)***	392.4 (66.2)***
Round3	1,108.5 (80.3)***	718.9 (64.0)***	438.6 (64.1)***	438.6 (66.2)***
GFrequency	-23.0 (29.3)	-33.2 (23.6)	-36.4 (19.0)*	-30.8 (21.4)
Awareness	-26.1 (75.1)	-37.3 (60.4)	55.6 (48.7)	-2.7 (54.8)
Need	142.6 (66.9)**	57.2 (53.9)	-11.2 (43.5)	62.8 (48.9)
PFrequency	11.1 (8.5)	9.1 (6.8)	6.6 (5.5)	9.0 (6.2)
PPrice	-45.8 (30.5)	-18.9 (24.6)	-38.8 (19.8)**	-34.5 (22.3)
Age	-18.8 (4.2)***	-15.4 (3.3)***	-11.6 (2.7)***	-15.3 (3.0)***
Education	-102.2 (55.9)*	-37.3 (44.9)	-30.4 (36.2)	-56.7 (40.8)
Household size	10.2 (48.5)	-21.0 (39.0)	-38.9 (31.5)	-16.7 (35.4)
Income	-0.1 (29.3)	10.1 (23.6)	7.9 (18.9)	6.0 (21.4)
Intercept	1,103.7 (506.7)**	926.1 (407.5)**	915.1 (329.4)***	892.8 (370.9)**
Good	—	—	—	114.1 (66.2)*
Super	—	—	—	153.4 (66.2)**
Good*Round2	—	—	—	300.2 (93.6)***
Good*Round3	—	—	—	280.3 (93.6)***
Super*Round2	—	—	—	703.7 (93.6)***
Super*Round3	—	—	—	669.9 (93.6)***
No. of Obs.	576	576	576	1,728
Log-likelihood	-4,729.9	-4,601.4	-4,560.7	-13,800.7

Note: *, **, and *** represent significance levels at 10%, 5%, and 1%, respectively.

from the rice grade premiums and included dummy variables for the “super” and “good” grades and their interactions with information rounds. Results from the pooled model are exhibited in column (4) of Table 7. The results also show that consumers add a premium for each rice grade. The quality grade for “super” has the largest premium, followed by the “good” and “normal” grades. The results also show that the biggest impacts of both grade-only and detailed grade information are on the “super” grade, followed by the “good” and “normal” grades.

Our results also show that the frequency at which rice grades are checked is negative in all three models; however, it is only significant for the “normal” grade. Specifically, consumers who frequently check rice grades are less likely to pay an additional premium for each rice grade. One possible explanation for this result is that these consumers may already know the price and quality of each grade. Participants who strongly desire a grading system are more willing to pay an additional premium for the “super” grade. Moreover, participants who purchase rice at a high price level are less willing to pay a premium for “normal” grade. Both elderly consumers and highly educated consumers have less incentive to pay an additional premium for each rice grade.

Figure 2 shows the cumulative distributions of the WTP differences between each rice grade and the “no test” rice across the three information rounds (we defined them as “quality premium”). Without any grade information, the distributions of the three premium grades are quite similar. Approximately 48% of the participants prefer the “no test” rice to the “normal” grade rice, and approximately 39% and 35% of participants prefer the “no test” rice to the “good” and “super” grade rice, respectively. However, after the provision of grade-only information, only about 17% of the participants prefer the “no test” rice to the “normal” grade rice; approximately 90% prefer the “good” grade rice to the “no test” rice; and almost all participants prefer the “super” grade rice to the “no test” rice. There are a few differences in the cumulative distributions of the premium for each rice grade between the grade-only and detailed grade information rounds (i.e., between rounds 2 and 3).

Objective 2: Consumers’ Valuation of a Mandatory Grading System

As previously discussed, we conducted a CV and IV study, using a DBDC format, to assess consumers’ valuation for a mandatory rice-grading system without a “no test” option. Table 8 shows the summary of the mean bid values provided in the CV and IV questions and participants’ responses to each of the questions.⁴ Given our use of the DBDC format, the mean bid provided to respondents in the first question in the DBDC in both valuation methods is approximately 1,102 KRW (approximately 1.2 CAD). For the first question in the DBDC, the “yes” response rate in the CV (20.7%) is slightly lower than the “yes” response rate in the IV (21.2%). The mean bids provided to respondents in the follow-up question (i.e., the second question in the DBDC) are approximately 783 KRW (approximately 0.87 CAD) and 753 KRW (approximately 0.84 CAD) in the CV and IV, respectively. Participants’ “yes” response rates in the CV and IV are approximately 43.3% and 44.7%, respectively.

⁴ We removed four participants from the analysis because they had not completed all the valuation questions.

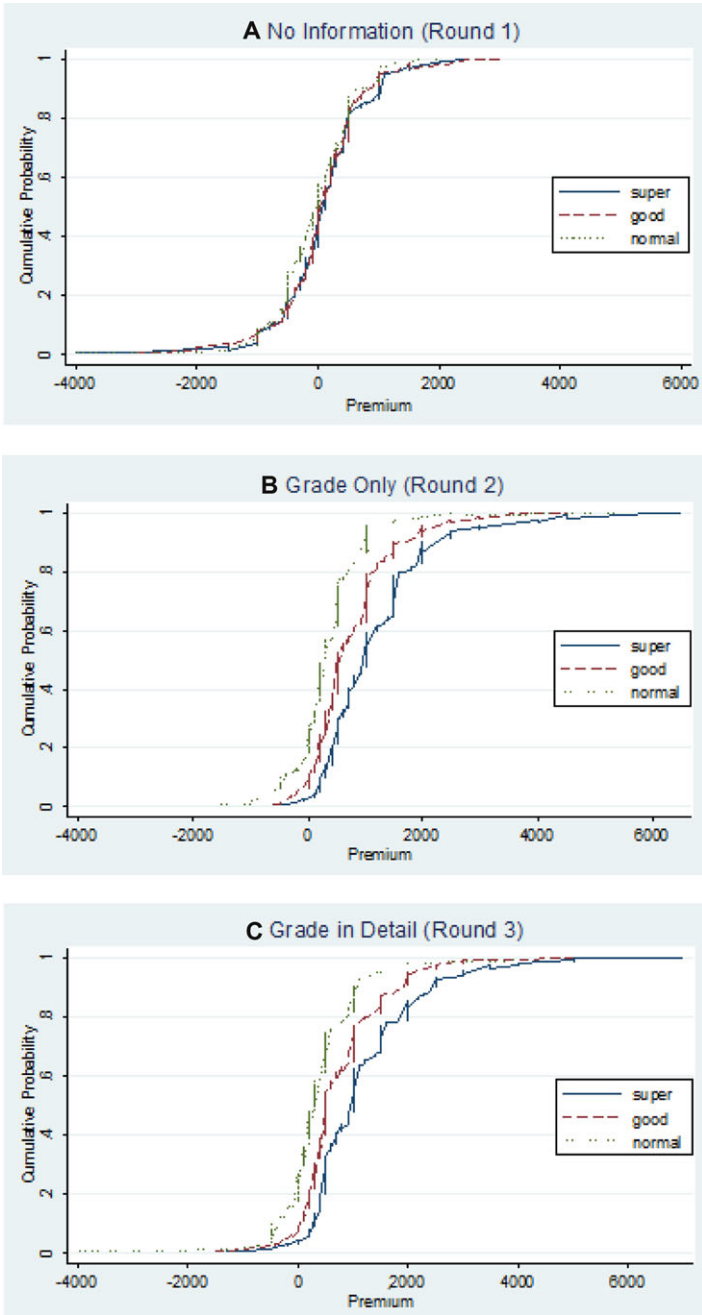


Figure 2. Cumulative distributions of the difference in WTP between rice grades and “no test”

Table 8. Average bid values provided and participants' responses in CV and IV ($N = 208$)

Method	Description	Mean	SD	Min.	Max.
CV	Bid value provided in the first question	1,101.9	576.2	200	2,000
	Bid value provided in the second question	783.2	639.2	100	3,600
	Response to the first question (1:Yes, 0:No)	0.21	0.41	0	1
	Response to the second question (1:Yes, 0:No)	0.43	0.49	0	1
IV	Bid value provided in the first question	1,101.9	574.8	200	2,000
	Bid value provided in the second question	752.9	533.6	100	3,600
	Response to the first question (1:Yes, 0:No)	0.21	0.41	0	1
	Response to the second question (1:Yes, 0:No)	0.45	0.49	0	1

Table 9. Regression results of consumer valuation for mandatory rice-grading system

Variables	Contingent valuation		Inferred valuation	
	Coefficient	Standard error	Coefficient	Standard error
Age	-12.9***	4.9	-6.7	4.2
Education	-176.7***	65.4	-70.3	57.3
Income	53.9	34.7	6.4	30.5
Household size	-63.7	56.6	-62.9	50.9
GFrequency	34.8	34.2	-11.9	30.3
Awareness	-0.4	87.7	103.9	77.1
Need	184.2**	79.4	93.5	70.6
PFrequency	-4.7	9.9	-6.0	9.1
PPrice	-66.3*	35.5	-54.6*	31.3
Intercept	1,486.5**	590.7	1,231.5**	515.2
Sigma	715.2***	54.7	622.0***	46.9
Log likelihood		-252.7		-236.4
No. of Obs.		192		192
Mean WTP (KRW/kg)		600.1***		623.4***

Note: Mean WTPs in the both valuation methods are estimated based on the regression results, and *** represents that mean WTPs are statistically different from zero at the 1% significance level. *, **, and *** in the regression result represent significance levels at 10%, 5%, and 1%, respectively.

Table 9 presents the regression results of participants' WTP for the mandatory rice-grading system without the "no test" option.⁵ Our results show that elderly consumers and highly educated consumers are less willing to pay for the mandatory grading system. Participants who agree to have a grading system on the basis of the consumers' right to information are more likely to pay a higher WTP for the mandatory grading system. Moreover, participants who pay a higher price for rice are less willing to pay for the mandatory rice-grading system. Prior to asking each valuation question, we provided participants with information about a possible price increase under the mandatory

⁵ We removed 16 participants from the regression analysis because they did not complete the questions used in the regression variables.

rice-grading system. It appears that participants who pay a high price for rice are more likely to avoid an additional price increase.

Based on the regression results, we estimate consumers' mean WTP for the mandatory rice-grading system without the "no test" option. The mean WTPs from the CV and IV are approximately 600 KRW (approximately 0.67 CAD) and 623 KRW (approximately 0.69 CAD), respectively, and these values are statistically different from zero. We also compare the WTP values from the two valuation methods to check the robustness of our results. According to the paired *t*-test, we fail to reject the equality of mean WTP between the two valuation methods (*t*-statistic: -1.25 , *p*-value: 0.21). Results indicate that rice consumers are willing to pay an approximate 15% premium above the average rice price (4,000 KRW) for the mandatory rice-grading system without the "no test" option, suggesting that rice consumers have a positive preference for the mandatory rice-grading system without the "no test" option.

We also investigate the sensitivity of the consumer welfare effects of implementing the mandatory rice-grading system without the "no test" option. We apply a framework that is similar to the one used by Lusk et al (2005) to estimate the change in consumer surplus. We assume that individuals consume one unit of a rice within a given period as shown by Giannakas and Fulton (2002). Under the current rice-grading system (i.e., the grading policy with the "no test" option), consumer *i* derives utility, $U_i - P^{DRWN} - c_i$, where U_i is consumer *i*'s maximum WTP for a unit of the rice, P^{DRWN} is the price of domestic rice under the rice-grading policy with the "no test" option, and c_i is individual *i*'s search cost for rice that has rice grade labels. A consumer also derives utility, $U_i - P^{IR} - \delta_i$, by consuming imported rice as a substitute for domestic rice,⁶ where P^{IR} is the price of imported rice and δ_i is individual *i*'s aversion to imported rice. In this regard, according to Lee et al (2015b), only approximately 1.9% of Korean consumers have positive attitudes toward the consumption of imported rice. We assume that $U_i - P^{DRWN} - c_i$ is greater than $U_i - P^{IR} - \delta_i$ because Korean consumers have a strong preference for domestic rice relative to imported rice (Lee et al 2014; Lee et al 2015b). Thus, consumer surplus under the current grading policy is given by the following equation:

$$CS_i^{DRWN} = \max \{ U_i - P^{DRWN} - c_i, U_i - P^{IR} - \delta_i \} = U_i - P^{DRWN} - c_i \quad (7)$$

If the mandatory rice-grading system without the "no test" option is implemented in Korea, consumer *i* derives utility, $U_i - P^{DRM}$, where P^{DRM} is the price of domestic rice under the mandatory rice-grading system without the "no test" option. We also assume that the utility of consuming domestic rice is greater than that of consuming imported rice. Thus, consumer surplus under the mandatory rice-grading system is defined as

$$CS_i^{DRM} = \max \{ U_i - P^{DRM}, U_i - P^{IR} - \delta_i \} = U_i - P^{DRM} \quad (8)$$

⁶ Even though the proportion of imported rice to the total rice consumption in Korea is small, the quantity of imported rice has been increasing since the opening up of the rice market through an MMA scheme during the 1995–2014 period. In addition, the change in the import framework to a tariffication system in 2015 has been a significantly sensitive issue in the Korean rice industry.

Table 10. Changes in consumer surplus relative to changes in domestic rice price under mandatory rice-grading system (N = 192, KRW/kg)

	CV				IV			
	5%	10%	15%	20%	5%	10%	15%	20%
Mean	363.9	210.9	−36.1	−236.1	379.7	227.9	−20.3	−220.3
Median	377.9	213.1	−22.1	−222.1	392.6	219.3	−7.4	−207.4
SD	316.4	282.2	316.4	316.4	235.6	172.3	235.6	235.6

Further, the change in consumer surplus from implementing the mandatory rice-grading system is specified:

$$\Delta CS_i^{DRM-DRWN} = CS_i^{DRM} - CS_i^{DRWN} \quad (9)$$

In order to estimate the change in consumer surplus relative to the changes in the price of domestic rice under the mandatory rice-grading system, we need to determine values for P^{DRWN} , P^{DRM} , and c_i . Consumers would decide on their WTP for the mandatory rice-grading system without the “no test” option, such that $U_i - P^{DRM} = U_i - P^{DRWN} - c_i$ because this is their dominant strategy in both valuation questions. Based on this equality, individual’s WTP is equal to c_i . We use the current average price of domestic rice for P^{DRWN} . We then increase the price of domestic rice under the mandatory rice-grading system (P^{DRM}) from 5% to 20%, with a gap of 5% relative to the current rice price (P^{DRWN}). The mandatory rice-grading system will raise domestic rice prices because of an increase in the processing cost of conducting the mandatory grading test.

Table 10 presents the changes in average consumer surplus from implementing the mandatory rice-grading system. Overall, welfare changes can be calculated by adding all the individual welfare changes. The results show that consumers gain benefits from the implementation of the mandatory rice-grading system without the “no test” option when the price increase is limited to 10% relative to the current rice price. However, on average, consumers suffer welfare losses if the rice price under the mandatory rice-grading system increases by more than 15% in both valuation methods.

CONCLUSIONS AND POLICY IMPLICATIONS

Differentiating domestic rice by quality grading is important since it can provide information to consumers about the relative quality of the rice products in the market, and thus, help them make informed purchasing decisions. The premium of quality-differentiated rice products could also provide an incentive to rice manufacturers to improve the quality of domestic rice. Consequently, the Korean government has implemented a rice-grading system so that consumers can identify the degree of quality of the rice products in the market. However, the current rice-grading system has been deemed ineffective because it allows the selling of rice that has not been tested for grading, and this decision to permit a “no test” option in the market has not provided a strong incentive to rice manufacturers to comply with the grading policy. To incentivize rice manufacturers to conduct the rice-grading test, quality premiums should explicitly be reflected in rice prices depending on

the quality grade of the products. Further, the costs of implementing the mandatory rice-grading system without a “no test” option should be partially supported by consumers, the primary beneficiaries of the system. In this study, we considered the feasibility of these changes in the current rice-grading system by (a) estimating consumers' premium for the rice quality grades (i.e., “super,” “good,” and “normal” grades) compared to the “no test” rice, (b) identifying the effect of rice-grading information on quality grade premiums, and (c) examining consumers' valuation of the mandatory rice-grading system without the “no test” option.

Our results from the experimental auction study showed that rice consumers cannot distinguish the quality of rice grades without rice-grading information. However, after they were given either rice grade-only or detailed grade information, they were willing to pay a 37% premium for the “super” grade, 23% premium for the “good” grade, and 11% premium for the “normal” grade compared with the “no test” rice. These results indicate that rice consumers value the provision of quality grades and are willing to pay more for higher than lower grade products, suggesting that rice-grading information is an important factor that allows consumers to differentiate each rice grade from “no test” rice and also among the different rice grades.

The results from our CV and IV study also suggest that consumers value a mandatory rice-grading system without a “no test” option. Specifically, they were willing to pay a 15% premium above the average market price of rice for the mandatory rice-grading system without the “no test” option. While we cannot definitively assess whether this 15% premium can cover the cost of implementing the mandatory rice-grading system without a “no test” option, it is an indication that consumers are willing to support a change in the grading system. On average, we also estimated that consumers would have welfare benefits from the mandatory rice-grading system when the price increase under the new policy is limited to 10–15% relative to the current rice price.

Overall, our findings generally suggest that consumers would value having quality grades in domestic rice in Korea. Specifically, consumers would value rice with quality grades, especially with “super” and “good” grades, more than rice without a quality grade (i.e., “no test” rice grade), and they also would prefer and value the mandatory rice-grading system without the “no test” option. Policy wise, this overall finding is a strong signal that the Korean government should adopt the mandatory rice-grade labeling system without the “no test” option. This revision in the current rice-grade labeling system will provide consumers more credible information to make better informed purchasing decisions and it could also provide an incentive to rice producers and processors to improve the quality of domestic rice.

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APPENDIX: CHEAP TALK SCRIPT

The experience from surveys is that people often state a higher WTP for an item than the amount they would truly be willing to pay for it in an actual purchasing situation. For instance, in a recent study, 80% of people said that they would buy a new food product, similar to those you will be asked about below, but when a grocery store actually stocked the product, only 43% of people actually bought the new product when they had to pay for it. This difference (43% vs. 80%) is what we refer to as hypothetical bias.

Accordingly, it is important that you respond to the following questions as if you were actually facing these exact choices in a store, that is, keeping in mind that paying more for a product means that you would have less money available for other purchases.

Source: Lee et al (2015a).